

Bryce Rogers, Ph.D, M.S. - Curriculum Vitae
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Summary

Computational and benchtop engineer with a background in biology, chemistry, and bioadjacent fields. **Scientific interest in creating and implementing informed systems for the betterment of biological, chemical or adjacent technology and study** | Ph.D in bioengineering: developed custom neural-network and analytic Python models optimized for behavioral bioplausibility in speech recognition and vision (*computational neurology/cognitive science*). Transparent, interpretable models with human-aligned architecture and behavior | M.S. in chemistry: protein semi-design (*biochemical/molecular-biological engineering*). DNA recombination, grafting, site-directed mutagenesis, cell culture, protein synthesis, characterization. 7+ years graduate-level quantitative life, chemical and adjacent sciences. Capable technical and lay communicator.

Education

Corvallis, OR / Remote	Oregon State University Ph.D Bioengineering Graduate Minor Biological Data Sciences GPA 3.95 <u>Summary</u> : Bioengineering core, focus on computational biological methods, data science/machine learning, and programming. Study and development of transparent, bioplausible models of word recognition and vision. <u>Course List</u> : Applied Machine Learning Advanced Computing of Biological Data Modern Statistical Methods for Large & Complex Datasets Data Science Tools & Programming Statistical Methods for Genomics Research Modeling of Physiological Systems Problem Solving in Biological Data Sciences Cell & Molecular Engineering Introduction to Data Science for Engineers Programming & Data Structures Foundations of Data Analytics Cancer Systems Biology Genotyping by Sequencing Drug & Medical Device Regulation in Technology Python I Python II Command Line Data Analysis (UNIX/Bash) Introduction to UNIX/Linux [Python, R, Bash, Julia, Matlab]	Sep 2020 - Mar 2025
Fort Collins, CO	Colorado State University M.S. Chemistry Chemical Biology Program GPA 3.52 <u>Summary</u> : Chemical biology core, focus on creating technology and/or therapeutics. Study and development of macro/molecular tools to control cellular biochemistry and treat human disease. <u>Course List</u> : Chemical Separations Molecular Genetics Chemical Biology I Chemical Biology II Biomolecular Engineering and Synthetic Biology	Aug 2013 - Sep 2016
Tacoma, WA	University of Puget Sound B.S. Biochemistry, Spanish Literary Studies GPA 3.05 <u>Summary</u> : Liberal arts core, biochemistry and language dual focus. <u>Course List</u> : Physical Biochemistry Organic Chemistry I Organic Chemistry II Metabolic Biochemistry Inorganic Chemistry Analytical Chemistry Physical Chemistry Genetics Cell Biology Ecology Natural Products Physics I Physics II Multivariate Calculus	Sep 2008 - May 2012

Research

Corvallis, OR /
Remote

Oregon State University

Sep 2020 - Mar 2025

Graduate Research Assistant (4 years, 7 months)

Advisor: Dr. Kevin Brown

Developed and characterized computational models at the interface of neuroanatomy/physiology, psycholinguistics, statistics, and programming. Bioplausible, customized and interpretable neural-network and analytic Predictive Coding (PC) models of speech recognition and vision (cf. Rao & Ballard, 1999). *Python, data science libraries (e.g. numpy, pandas, and matplotlib), HPC. Model workflow: architecture/planning, exploratory statistics, input preprocessing, training/optimization, evaluation/analysis, visualization/reporting.* First-author publication, Neural Computation, 2026 (in press): first known formal, recurrent PC neural-network classifier of over-time, speech-like input with benchmarked behavioral realism (lexical competition) in human spoken word recognition.

Co-first-author manuscript in preparation: analysis and extension of analytic PC and signal-sharpening models for bioplausibility in lexical competition and toward falsification of a purportedly diagnostic signal of neural prediction error (cf. Blank & Davis, 2016). ~ \$30,000 in academic entrance or performance awards from OSU/affiliates. **Python | Statistical/Mathematical Modeling | Machine Learning/Deep Learning/Data Science | Computational Neurology/Cog. Science | Git/Github Version Control | HPC**

Fort Collins, CO

Colorado State University

Mar 2014 - Sep 2016

Graduate Research Assistant (2 years, 6 months)

Advisor: Dr. Brian McNaughton

Engineered (resurfaced/grafted “semi-design”) protein therapeutic/technologic candidates for inhibition of disease-relevant protein-protein interactions. DNA *library creation and expression (recombination, yeast display), high-throughput screening (FACS), and biochemical characterization (e.g. gel, column chromatography, ELISA assay, isothermal titration calorimetry)*. Maturation of *high-affinity binders* of Ankyrin Repeat (AR) targets; *screening* for synthetic AR proteases. Second-author publication, Biochemistry, 2014: characterizing the binding interface between oncoprotein Gankyrin (AR) and a grafted ATPase. **Molecular Biology | Biochemistry | Protein Semi-Design**

Colorado State University

Jan 2014 - Mar 2014

Research Rotation (1.5 months)

Advisor: Dr. Brian McNaughton

Began solubilizing the Ankyrin Repeat (AR)-binding domain of S6-ATPase to study the binding interaction between S6-ATPase and AR oncoprotein Gankyrin. The AR-binding domain of S6-ATPase was recombinantly grafted onto structurally-homologous and soluble protein, FtSH. The resulting grafted *FtSH-S6* exhibited high affinity for Gankyrin and was used to *quantify Gankyrin-S6 binding thermodynamics*. **Molecular Biology | Biochemistry | Protein Semi-Design**

Colorado State University

Oct 2013 - Dec 2013

Research Rotation (1.5 months)

Advisor: Dr. Robert Cohen

Synthesis of high-affinity protease-controlled inhibitor of K63-linked polyubiquitin to *study the role of K63-linked pUb in autophagy*. *Synthesized chemical tag of Ub-E2/E3 thioesters* to separate biochemically-desirable (n)Ub chains from lysate. **Molecular Biology | Biochemistry | Synthesis**

Colorado State University

Sep 2013 - Oct 2013

Research Rotation (1.5 months)

Advisor: Dr. Alan Kennan and Dr. Chris Snow

Used HippDB (*database of protein-protein interactions* with one or more α -helices in binding interface) to search for protein scaffolds with structural potential to inhibit Ebola C-peptide (mandatory structure for viral membrane fusion, infection). Best candidate scaffolds *combinatorially optimized at the amino acid/rotamer level* using computational Python platform SHARPEN (Systematic Hierarchical Algorithms for Rotamers and Proteins on an Extended Network), estimating low energy states. *Molecular docking simulations between scaffold candidates and C-peptide as screening for optimal scaffolds* of C-peptide helical inhibitor grafting. **Computational Synthetic Biology | Bioinformatics**

Tacoma, WA

University of Puget Sound

Aug 2011 - Dec 2011

Directed Research (3.5 months)

Advisor: Dr. John Hanson

Synthesized acetylated maltose analogs in order to study the active site and maltose-cleaving mechanism of the enzyme MalA. **Organic Chemical Synthesis**

Teaching

Corvallis, OR /

Remote

Oregon State University

Sep 2021 - Mar 2024

Graduate Teaching Assistant (3 quarters)

Undergraduate Chem Eng., Molecular Bio. & Biochem.: *1 quarter Material Balances & Stoichiometry, 1 quarter Energy Balances, 1 quarter Biochemistry & Molecular Biology laboratory* | **Molecular Biology & Biochemistry**: e.g., transformation, sequencing | **Chemical Engineering**: e.g., material and energy balances, flow diagrams | Teaching: tutorials, safety, digital communication, grading

Fort Collins, CO

Colorado State University

Aug 2013 - Aug 2016

Graduate Teaching Assistant (7 semesters)

Undergraduate Chemistry: *2 semesters General Chemistry laboratory, 5 semesters Organic Chemistry laboratory* | **Organic Chemical Synthesis**: e.g., **reflux, distillation, extraction, recrystallization, TLC, column chromatography, H-NMR, IR, polarimetry** | Teaching: tutorials, safety, prep. of lectures and quizzes, office hours, digital communication, grading

Publications & Writing

- “Extending a formal model of predictive coding to human spoken word recognition” 2026
Bryce Rogers, Monica Yin-Chen Li, Thomas Hannagan, James S. Magnuson, and Kevin Brown
Neural Computation, 2026 (in press, MIT Press)
- “Predictive coding models in human spoken word recognition” 2025
Bryce Rogers, incl. collaborative research
Oregon State University, 2025 (dissertation) ([link](#))
- “Characterization of the Binding Interaction between the Oncoprotein Gankyrin and a Grafted S6 ATPase” 2014
Alex M. Chapman, Bryce E. Rogers, and Brian R. McNaughton
Biochemistry, 2014 ([link](#))
- “On prediction error and signal-sharpening models in word recognition” NA
[Phoebe Gaston, Bryce Rogers]*, Kyra Kadhim, Kevin Brown, and James S. Magnuson [*co-first authors]
In preparation for *PLOS Biology* (est. submission 2026/27)

Academic Awards (>\$105K outside of graduate assistantships)

- Oregon State University* College of Engineering Dean’s Award 2022
Graduate merit scholarship from OSU College of Engineering | ~ \$3,500 for academic performance
- Oregon State University* Jean and Richard Josephson ARCS Foundation Scholar 2020
Graduate entrance scholarship from Achievement Rewards for College Scientists | \$18,000 for previous academic performance
- Oregon State University* Provost Distinguished Graduate Scholar 2020
Graduate entrance scholarship from OSU Bioengineering | \$7,500 for previous academic performance
- University of Puget Sound* Trustee Scholarship 2008
Undergraduate entrance scholarship from University of Puget Sound | \$80,000 stipend for previous academic performance

Formal Communication

- Achievement Rewards for College Scientists (ARCS) Foundation Annual Scholar Event* 2022
“Predictive Coding Models in Human Cognition” Poster | Portland, OR
- Oregon Bioengineering Symposium* 2022
“Predictive Coding Models in Human Cognition” Poster | Remote / Corvallis, OR
- Oregon Bioengineering Symposium* 2021
“Formal Predictive Coding Models in Human Vision and Beyond” Poster | Remote